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Designing Clinical Decision Support for Early Cochlear Implant Candidacy Identification in Medical Practice

A Qualitative Interpretative Single Case Study

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Abstract. *Despite growing evidence that cochlear implants (CIs) can significantly improve speech understanding and quality of life across an increasingly broad range of patient groups, many potentially suitable candidates remain unidentified. Current referral pathways are often shaped by limited assessment procedures and uncertainty regarding candidacy criteria, contributing to persistent under-referral in hearing healthcare. Against this background, this thesis explores how a self-administered, language-agnostic psychoacoustic hearing screening tool can be designed to support clinicians in the early identification of potential CI candidates. Grounded in a qualitative, interpretative single case study design, the research investigates how psychoacoustic screening results can be translated into a clinical decision support (CDS) concept. The study combines insights from psychoacoustic research, CDS literature, empirical findings and regulatory considerations surrounding digital health technologies. The findings highlight that clinicians require decision-support tools that provide rapid, transparent and workflow-compatible orientation. These insights informed the development of a non-directive CDS concept centered around layered visualization approaches, progressive disclosure and flexible interaction principles that allow clinicians to explore psychoacoustic listening profiles in greater depth.*

Keywords: cochlear implant candidacy, clinical decision support, service design

1 Introduction

Hearing is fundamental to human connection. It shapes how people communicate, participate in society and experience everyday life. When hearing deteriorates, the consequences extend far beyond reduced auditory perception, often affecting social inclusion, mental well-being and overall quality of life (Blazer & Tucci, 2019; Rutherford et al., 2018). For individuals with severe hearing loss, cochlear implants (CIs) represent one of the most significant technological advances in modern medicine. They provide a reliable and well-established solution when conventional hearing aids (HAs) no longer deliver sufficient benefit. Despite their proven effectiveness, however, CIs remain substantially underutilized worldwide (Marinelli & Carlson, 2021; Sorkin & Buchman, 2016). Rather than reflecting limitations in available solutions, a major barrier to CI uptake lies in the early stages of the care pathway, where potential candidates are either not referred or referred too late for further evaluation (Neukam et al., 2024). Healthcare professionals may lack the tools, knowledge or resources required to assess whether a patient would benefit from a CI, particularly when functional testing is not routinely conducted outside specialized settings. At the same time, non-specialist professionals and patients themselves often have limited awareness of CIs as a treatment option. These challenges contribute to missed opportunities for timely intervention and prolong the use of HAs even when they are no longer effective (Ebrahimi-Madiseh et al., 2020; Holder et al., 2018).

Even in healthcare systems where cochlear implantation is accessible and publicly funded, the process of candidacy identification remains complicated. Although extensive research has established eligibility criteria, these have not translated into a streamlined or standardized process in practice (Luntz et al., 2018). Furthermore, many current tests depend on language-specific stimuli or require clinical supervision, limiting their accessibility and scalability. While recent efforts have explored language-agnostic and self-administered approaches, there is still no widely adopted, standardized solution. This highlights a critical gap: the absence of reliable, user-friendly tools that support clinicians in identifying potential CI candidates (Ching et al., 2022; Ebrahimi-Madiseh et al., 2020; Govindan et al., 2025; Patro et al., 2023). Addressing this gap requires more than incremental improvements in clinical practice. It calls for a rethinking of how candidate identification is supported within the broader healthcare ecosystem. In particular, there is an opportunity to design tools that not only assist clinicians in decision-making but also fit seamlessly into existing workflows and constraints. Such tools must balance clinical rigor with usability, while also aligning with the regulatory and practical realities of healthcare delivery.

Against this backdrop, this thesis builds on an earlier internship project conducted in collaboration with MED-EL to explore whether a previously unvalidated self-administered screening concept could support clinicians in the earlier and more effective identification of potential CI candidates. While MED-EL focuses on developing the underlying test, this thesis investigates how such a solution can be designed and integrated as a service within clinical practice. Specifically, it addresses the following research question:

How can a self-administered hearing screening tool be designed to support clinicians in the early identification of potential cochlear implant candidates?

2 Literature Review

Hearing loss is often quantified in decibels, yet its true consequences extend far beyond reduced audibility, unfolding in restricted participation in everyday life and compromised psychological well-being (Blazer & Tucci, 2019). Persistent communication barriers, the constant cognitive effort required to decode sound and associated social withdrawal can foster feelings of loneliness, anxiety and depression, and over time may contribute to accelerated cognitive decline and an increased risk of dementia (Livingston et al., 2017; Rutherford et al., 2018; Shankar et al., 2013). Importantly, such consequences are not confined to profound deafness. Historically, milder forms of hearing loss were underestimated, based on the assumption that partial hearing was sufficient to guarantee normal auditory functioning. This perception is further reflected in findings showing that individuals with self-reported mild hearing loss were often advised to postpone intervention and monitor their hearing over time, or were informed that HAs would offer limited benefit – thereby reinforcing the view that less severe impairment does not require immediate treatment (EHIMA, 2012; Giolas & Wark, 1967; Kochkin, 2012). Yet growing evidence challenges this view, demonstrating that even unilateral or mild to moderate hearing loss is associated not only with increased psychological distress and social difficulties compared to individuals with normal auditory function but also with potential impacts on the development and maintenance of cognitive abilities, disadvantages in educational achievement and employment outcomes as well as economic hardship (Emmett & Francis, 2015; Idstad & Engdahl, 2019; Järvelin et al., 1997; Lieu, 2018; Monzani et al., 2008). These findings underscore a broader insight: Hearing loss – even when seemingly limited – can lead to a marked decline in quality of life (Sladen et al., 2018).

HAs are generally regarded as the primary, non-invasive intervention for hearing loss and effectively support many individuals with mild to moderate hearing loss in everyday situations (Ferguson et al., 2017). However, measurable hearing loss alone does not automatically make someone an appropriate HA candidate, and a notable proportion of users fitted with HAs report insufficient benefit and discontinue use when expected improvements are not adequately met (Kochkin, 2000; McCormack & Fortnum, 2013). Evidence from several studies indicates that user dissatisfaction commonly stems from reduced speech clarity, a perceived lack of meaningful improvement in everyday listening situations characteristic of real-world social environments and poor sound quality (Bertoli et al., 2009; Hong et al., 2014; McCormack & Fortnum, 2013). This limitation reflects the underlying mechanism of HAs: While they enhance audibility by amplifying incoming sounds and compensating for reduced sensitivity associated with outer hair cell damage – thereby increasing access to softer sounds – amplification alone does not restore normal auditory perception and often remains insufficient for understanding speech in noise (Moore, 2026). Because HAs amplify all acoustic input rather than selectively enhancing speech, users may continue to experience communication difficulties, particularly in noisy environments (Picou et al., 2013).

In clinical practice, individuals who continue to struggle with hearing difficulty with poor speech understanding despite appropriately fitted amplification often meet criteria for cochlear implantation (Marinelli & Carlson, 2021; Tsai Do et al., 2024). By converting sound into electrical signals that directly stimulate the auditory nerve, CIs compensate for non-functioning hair cells and present a promising solution for individuals with sensorineural hearing loss who gain limited or no benefit from conventional HAs (Choi et al., 2016; Gaylor et al., 2013). Widely established as the most suitable intervention for severe-to-profound bilateral hearing loss and poor aided speech recognition, growing evidence indicates that even individuals with greater residual hearing, asymmetric hearing loss or unilateral deafness – where one ear retains normal function – also

experience significant improvements in speech recognition and self-perceived health-related quality of life, referring to an individual's perceived mental and physical well-being (Arndt et al., 2017; Firszt et al., 2012; Gaylor et al., 2013; Sladen et al., 2018; Távora-Vieira et al., 2013). Notably, improvements in health-related quality of life may even exceed those observed in speech recognition abilities, underscoring the substantial psychosocial impact of hearing implants beyond purely auditory outcomes (McRackan, Bauschard, Hatch, Droghini, et al., 2018; McRackan, Bauschard, Hatch, Franko-Tobin, et al., 2018; Sladen et al., 2018).

2.1 Barriers to Timely Identification and Referral of Cochlear Implant Candidates

Although candidacy criteria continue to expand alongside growing evidence of benefit in broader and previously nontraditional populations, the translation of this evidence into timely identification and uptake remains insufficient – leaving many qualified candidates untreated (Eikelboom et al., 2025; Looi et al., 2017; Sorkin, 2013; Távora-Vieira et al., 2013). While this is partly explained by patient-related factors such as sociodemographic disparities, economic factors or concerns about surgery, the persistent gap between clinical eligibility and implantation is increasingly understood as a consequence of structural barriers within the hearing-care system (Bierbaum et al., 2020; Greiner et al., 2023; Kaufman et al., 2026; Mahendran et al., 2022; Mangan et al., 2022; Swords et al., 2024; Thompson et al., 2022; Zhao et al., 2025). A central barrier lies within the referral pathway itself. The transition from initial clinical contact to specialized CI evaluation represents a critical bottleneck in the care pathway (Looi et al., 2017). Because healthcare professionals often serve as the first point of contact for individuals with hearing difficulties, their knowledge, confidence and communication skills strongly influence whether potential candidates are identified and referred for further evaluation (Ebrahimi-Madiseh et al., 2020; Ramos-Macías et al., 2023; Reddy et al., 2022). However, clinicians' limited familiarity with CI technology, unclear candidacy criteria and referral indicators as well as a lack of confidence when discussing implants with patients have repeatedly been identified as a barriers to appropriate referral (Le et al., 2026; Vickers et al., 2016).

One important explanation for these knowledge gaps is the limited integration of cochlear implantation into medical education. Cochlear implantation remains insufficiently represented in tertiary training, and opportunities for continued professional development throughout clinicians' careers are often limited (Ben-Itzhak et al., 2005; Bierbaum et al., 2020; Raine et al., 2016; Sorkin, 2013). As a result, many clinicians remain uncertain about expected outcomes and eligibility criteria, which can undermine confidence when discussing implants with patients or initiating referrals (Chundu & Buhagiar, 2013; Mathews et al., 2009). Importantly, these challenges are not confined to primary care physicians. Knowledge gaps regarding current CI candidacy and outcomes following implantation have also been reported among hearing care professionals such as audiologists and HA dispensers (Chundu & Buhagiar, 2013; Sorkin, 2013). This may result in referral delays, under-identification of appropriate candidates and the continued use of HAs despite diminishing suitability (Sharma et al., 2023; Sorkin & Buchman, 2016). Reflecting this uncertainty, Reddy et al. (2022) found that a substantial proportion of non-otolaryngologists indicated they would refer patients with any degree of hearing loss for cochlear implantation, highlighting limited familiarity with current candidacy guidelines. Compounding this issue, cochlear implantation lacks the widely adopted referral and management standards that guide clinical decision-making in many other areas of medicine. This lack of clear, widely recognized guidelines continues to hinder its consistent recognition and implementation as a standard treatment for hearing loss (Sorkin, 2013). In addition, organizational incentives in some hearing-care settings, such as a focus on HA sales

rather than patient outcomes, may indirectly discourage referral for CI evaluation. Alongside unclear referral guidelines and fragmented care coordination, this can slow progression through the care pathway and delay the identification of eligible candidates (Le et al., 2026).

Referral decisions may also be shaped by subjective assumptions about who constitutes an appropriate candidate. Hearing loss is sometimes given lower clinical priority compared to other health conditions, which may reduce the likelihood that clinicians initiate discussions about rehabilitation or cochlear implantation (Le et al., 2026). In practice, clinicians may also hesitate to refer patients due to factors such as advanced age, perceived lack of social support, cultural identification with the Deaf community or doubts about a patient's ability to engage in rehabilitation (Cullington et al., 2024; Looi et al., 2017). Yet these considerations do not determine CI candidacy. Eligibility is primarily defined by audiological indicators, most importantly the degree of hearing loss and aided speech recognition performance. While the final determination of candidacy is made by CI specialists through a comprehensive assessment that also considers factors such as health status, lifestyle, motivation and psychosocial factors, the initial referral decision should rely primarily on objective audiological findings (Leigh et al., 2016; Zwolan et al., 2020). When subjective judgments begin to outweigh these objective criteria, potential candidates risk being filtered out before they ever reach specialist evaluation, reinforcing systemic delays in access to cochlear implantation (Huart & Sammeth, 2009).

At the same time, structural limitations in clinical testing environments can further complicate the identification of suitable candidates. Routine audiological assessments typically rely on pure-tone thresholds and speech recognition tests performed under earphones in quiet conditions (Gifford, 2011). While these measures provide valuable diagnostic information, they do not necessarily reflect real-world listening performance with amplification (Goehring et al., 2017). CI candidacy, however, is ideally determined using best-aided speech recognition testing, where speech understanding is measured in each ear with optimally fitted HAs under calibrated conditions. Such assessments require specialized equipment and testing environments and therefore cannot be routinely performed in many clinical settings (Cullington et al., 2024; Zeitler et al., 2024). Consequently, clinicians often rely solely on the audiogram when making referral decisions. However, pure-tone thresholds alone offer only limited information about how effectively a patient can communicate with properly fitted HAs at typical conversational listening levels and overestimate real-world aided performance, meaning that individuals who could benefit from cochlear implantation may be incorrectly assumed to achieve sufficient benefit from amplification alone (Chundu & Flynn, 2014; Gifford, 2011; McRackan, Fabie, et al., 2018). These diagnostic limitations have become increasingly consequential as CI candidacy criteria have expanded. In the past, candidates were typically profoundly deaf individuals with very limited speech recognition, making them relatively easy to identify using standard audiological tests (McRackan, Fabie, et al., 2018). Today, however, many implant recipients retain considerable residual hearing and measurable speech perception, which makes accurate identification more complex and requires more specialized testing procedures (Carlson et al., 2018).

Overall, these diagnostic, organizational and perceptual barriers can hinder timely identification of suitable CI candidates. The literature suggests that under-referral is less a question of clinical effectiveness than of systemic barriers, highlighting the need for improved clinician awareness, clearer referral pathways and stronger coordination within the hearing-care system.

2.2 Current Candidacy Assessment and Opportunities for Clinical Decision Support

While the previous chapters highlighted factors contributing to the gap between candidacy identification and implantation, as well as barriers within clinical referral pathways, translating these insights into practical improvements remains challenging. Although CI candidacy is ideally evaluated under best-aided listening conditions – the gold standard for determining the limits of HA benefit – implementing the full diagnostic protocol can be demanding in everyday clinical practice, particularly when referral decisions must be made under conditions of limited time, incomplete information and constrained diagnostic resources (Choi et al., 2016). Establishing a best-aided condition requires careful HA fitting and verification before speech perception testing can be performed. Depending on the patient, this process may involve additional steps such as producing earmolds or allowing time for HA adaptation, and even experienced HA users often require further assessment to ensure an appropriate fit. As a result, confirming the best-aided listening condition can require considerable time and clinical effort. Speech-based assessments introduce further limitations: Test materials are often available only in a limited number of languages, which may disadvantage patients whose native language differs from the available stimuli and complicate comparisons across clinical settings. In addition, the restricted number of sentence lists can introduce learning effects when patients undergo repeated evaluations (Drennan et al., 2014; Landsberger et al., 2019; Shim et al., 2014).

These practical constraints have motivated research into additional measures to support the identification of potential CI candidates, especially at earlier stages of the referral pathway. While CIs have demonstrated strong outcomes in restoring speech recognition in quiet, one key challenge for many CI users is reduced speech perception in noisy listening environments (Dornhoffer et al., 2021; Won et al., 2007; Zaltz et al., 2020). Variability in CI outcomes is therefore not primarily explained by the audibility of speech cues alone. Speech processors transform acoustic signals into patterns of electrical stimulation across the electrode array, meaning that conversational-level speech is typically audible when device settings are appropriately adjusted (Henry et al., 2005). Instead, because speech signals are characterized by complex spectral and temporal modulations, differences in performance largely depend on how effectively listeners can extract speech cues from these patterns of electrical stimulation (Ankmnal Veeranna et al., 2019; Apoux et al., 2013). Successful speech recognition therefore partly depends on the ability to perceive the patterns defining different speech sounds, especially the identification of spectral peaks that carry important phonetic information (Henry et al., 2005; Won et al., 2007). Consistent with this, empirical findings indicate that tests of spectrotemporal modulation testing, evaluating the ability to distinguish frequency differences (spectral) and timing changes (temporal) in sounds, are closely associated with aided speech perception performance, suggesting that they may serve as a useful indicator when evaluating CI candidacy (Shim et al., 2014; Van Groesen et al., 2023; Won et al., 2007, 2010, 2011). Beyond their predictive value, unaided non-linguistic psychoacoustic approaches like this offer several practical advantages making them attractive as supportive screening measures. Because these tasks rely on speech-agnostic stimuli, they are largely independent of language and less influenced by cognitive or educational factors that may affect speech recognition outcomes (Van Groesen et al., 2023). In addition, they can often be administered more efficiently in terms of both time and cost than traditional protocols that require extensive HA fitting procedures and prolonged speech perception testing, which makes them particularly appealing in clinical settings where resources for full HA trials or extended speech testing are limited (Shim et al., 2014).

Moreover, non-linguistic psychoacoustic tests show minimal acclimatization, reducing the learning effects that can occur with repeated speech-based tests (De Jong et al., 2018; Won et al., 2007). Despite these promising findings, psychoacoustic approaches have so far remained largely confined to research settings, and clinically usable screening tools based on these principles are still limited (Landsberger et al., 2019). Despite increasing awareness of under-referral for cochlear implantation, many of the barriers identified in the literature represent systemic challenges that require substantial time, coordination and structural change to address. Consequently, there is growing recognition that clinicians may benefit from practical forms of clinical decision support (CDS) that assist in identifying potential CI candidates earlier in the care pathway. This gap between promising psychoacoustic research and practical clinical implementation highlights an opportunity to design a non-diagnostic CDS that can assist clinicians during routine consultations by providing rapid indications of whether a patient might benefit from further CI evaluation.

2.3 Design Principles for Effective Clinical Decision Support

A CDS system is defined as "software that is designed to be a direct aid to clinical decision-making, in which the characteristics of an individual patient are matched to a computerized clinical knowledge base and patient-specific assessments or recommendations are then presented to the clinician for a decision" (Sim et al., 2001). Particularly in settings with limited resources, CDS tools offer a comparatively efficient way to support clinical decision-making and improve outcomes. As organizations consider how to allocate resources within health IT, evidence suggests that implementing CDS can provide substantial benefits across a range of clinical processes early on (Bayor et al., 2025; Feldman et al., 2018). In addition, their use has been associated with enhanced clinician performance, greater adherence to clinical guidelines, a higher proportion of patients receiving desired care and reduced medical errors (Kwan et al., 2020; Prgomet et al., 2017; Taheri Moghadam et al., 2021). Despite these benefits, the adoption of CDS in clinical practice remains relatively low, with several barriers hindering their integration into routine care (Kouri et al., 2022). Barriers include a lack of knowledge or confidence among clinicians, the perception of reduced professional autonomy, disruptions to established workflows and, importantly, issues related to usability and system design (Can Cubukcu, 2025; Devaraj et al., 2014; Liberati et al., 2017; Miller et al., 2018). In this context, the design of CDS emerges as a critical factor influencing their effectiveness, usability and adoption (Babione et al., 2020; Bayor et al., 2025). The way information is presented, how users interact with the system and how well it aligns with clinical workflows all shape user experience and perceived usefulness. Given the complexity of healthcare environments, the diversity of users and a lack of evidence-based guidelines, designing CDS user interfaces that meet these requirements is a significant challenge (Bayor et al., 2025; Miller et al., 2018). To address these challenges, user-centered design (UCD) approaches have been increasingly emphasized. UCD focuses on designing systems around the needs, behaviors and contexts of their users. In the context of CDS, this involves actively engaging clinicians and patients throughout the design, development and integration process to ensure that the final solution integrates into their clinical workflows and aligns with their needs and expectations (Babione et al., 2020; Bayor et al., 2025). Drawing on a narrative review that examined and consolidated design features linked to successful CDS, three interrelated design dimensions can be identified: interface, information and interaction (Miller et al., 2018).

The interface design plays a central role in how effectively CDS is perceived and used in practice. Across the literature, visualization emerges as a key design mechanism for CDS, enabling clinicians to efficiently interpret complex patient data and reduce cognitive load. Rather than relying on text-

heavy interfaces, it is recommended to use a combination of graphical representations – such as graphs, tables, buttons, iconic language and scroll bars – to achieve an appropriate information density. Graphs support rapid sense-making by making changes over time visible. In particular, line graphs are consistently preferred over bar or column charts for identifying trends, patterns and deviations in patient data. Tables remain valuable when precise values are required. Consequently, a combination of both formats is recommended, enabling clinicians to switch between overview and detail as needed. Color-coding further enhances interpretability by directing attention to clinically relevant states, with traffic-light schemes (e.g. red for high risk) widely used due to their intuitive association with urgency (Jayawardena et al., 2025; Miller et al., 2018). Visual cues should be used consistently across systems to ensure that their meaning is immediately recognizable. Consistency in visual signaling allows clinicians to rely on rapid perceptual recognition rather than requiring additional cognitive effort to interpret differences, thereby supporting more efficient and accurate decision-making (Horsky et al., 2012). Beyond visualization, interface design should prioritize simplicity. Techniques such as concise wording, consistent terminology, clear visual hierarchy and 'at-a-glance' layouts support efficient interpretation in time-constrained environments while features such as expandable and collapsible elements allow users to access additional detail as needed without overwhelming the initial display (Casey et al., 2023; Miller et al., 2018; Tsopra et al., 2014). At the same time, emerging interface concepts – such as collaborative social media-like features – offer opportunities to support knowledge exchange and peer learning (Bayor et al., 2025).

The design of information within CDS systems is closely tied to issues of trust, clarity and clinical relevance. Information must be presented in a clean, concise and unambiguous manner using standardized terminology to avoid misinterpretation. For example, if alerts are classified into defined severity levels, the same terms should be applied consistently across all alerts, messages and textual references (Horsky et al., 2012). Ambiguity in language or unfamiliar terminology can lead to incorrect application of recommendations. At the same time, CDS should go beyond presenting raw data by offering contextually relevant, evidence-based recommendations. These recommendations should be transparent and justified, for example by providing underlying reasoning, references to clinical evidence or access to additional resources (Horsky et al., 2012; Miller et al., 2018; Tsopra et al., 2014). If recommendations conflict with clinicians' experience or lack clear justification, they are more likely to be disregarded. Systems should be designed to avoid appearing as opaque 'black boxes'. Instead, the underlying reasoning and calculation values should be accessible when needed, allowing users to understand, verify and critically assess the basis of the system's outputs (Horsky et al., 2012). However, a critical distinction must be maintained between supportive and directive communication. While directive recommendations may appear efficient, they risk undermining clinician autonomy and may be rejected, particularly by experienced users. Instead, CDS should offer credible suggestions and supporting information that enrich decision-making without prescribing actions and allow for adjustment of responses to accommodate nuances beyond binary parameters (Babione et al., 2020).

The design of interaction determines how seamlessly CDS integrates into clinical workflows and how much effort is required from users. Mismatches between when information is required and when it becomes available can render CDS ineffective or disruptive. A key requirement is therefore efficiency: Systems must deliver timely recommendations at the point of decision-making while minimizing the need for manual input and interaction with the system. Excessive data entry, particularly when relying on free-text input, has been identified as a major barrier to adoption. Automated data integration and interoperability with existing systems, such as electronic health records (EHRs), are therefore critical to enabling accurate and timely decision support (Bayor et al.,

2025; Jaspers et al., 2011; Jayawardena et al., 2025; Mastrianni et al., 2023; Miller et al., 2018). Although stand-alone systems may be easier to implement, poor integration can increase cognitive load, disrupt clinical workflows and lead to incomplete or delayed data capture (Sutton et al., 2020; Wright & Sittig, 2008). Interaction design should also minimize cognitive load by reducing the number of steps, screens and interactions required, for example through the use of structured input fields, dropdown menus and streamlined navigation. Additional interaction principles include flexibility and error tolerance, allowing users to modify inputs, correct mistakes and adapt the system to changing clinical situations (Miller et al., 2018; Tsopra et al., 2014). Features such as automated alerts and prompts can enhance usability but must be carefully designed to avoid alert fatigue (Jayawardena et al., 2025). Evidence from a study examining physicians' perceptions of a novel CDS intervention suggests that clinicians differ in when they perceive CDS as most useful throughout the clinical encounter, with some valuing it early in the visit, others during diagnostic and treatment decisions and others at later stages. However, the study also indicates that early, passive prompts – encouraging but not mandating interaction – may increase the likelihood of CDS use (Casey et al., 2023).

3 Method

This study adopts a qualitative, interpretative single case study design to explore clinical decision-making and investigate how psychoacoustic screening results could be translated into a clinician-facing CDS concept. Accordingly, the study is structured around three interconnected components: a literature review, an empirical investigation and the interpretative translation of the findings into a clinician-facing CDS prototype. The literature review established the theoretical and contextual foundation of the study, informing the empirical investigation and subsequently guiding the development of the prototype. This methodological approach was grounded in both the exploratory nature of the research problem and the practical constraints and opportunities of the study context. A qualitative approach was chosen as it enables an in-depth understanding of how clinicians perceive and navigate complex referral decisions (Clark et al., 2021). It captures experience-based insights, allowing patterns, challenges and unmet needs to emerge that are difficult to identify through quantitative methods. Given the exploratory nature of this study and the limited existing knowledge on CI referral decision-making, this approach provides the necessary depth to inform design. Grounded in a constructivist research paradigm, the study approaches knowledge as something shaped through subjective perspectives and therefore seeks to understand how individuals make sense of their experiences rather than providing objective explanations (Yin, 2014). Accordingly, clinicians' responses are interpreted as context-dependent insights, with findings co-constructed through the interaction between participant input and researcher interpretation. Rather than representing objective truths, the results are used to derive design implications for the proposed solution (Baxter & Jack, 2015; R. Ponelis, 2015). The decision to adopt a single case study design focusing on individuals involved in CI referral processes reflects both practical constraints, including the limited timeframe of the study, and the opportunity to access a highly relevant participant group through the collaboration with MED-EL. Beyond feasibility, the case study serves as a bridge between research and practice. It builds on a previously developed prototype that has not yet been empirically validated and provides the foundation for its refinement. The insights generated aim to inform the iterative development of CDS for CI referral and may support future research that uses more collaborative, iterative approaches.

3.1 Secondary Data Collection

The literature review was conducted using a forward and backward search strategy following the concept-centric approach for information systems (IS) research outlined by Webster and Watson (2002). Based on the research question, a systematic search was performed in PubMed, selected for its strong focus on medical and clinical research and its relevance to healthcare-specific design constraints (see Figure 1). Two complementary search strings were applied. The first search string – *"cochlear implant*" AND ((candidacy OR "candidate identification" OR referral) AND (barrier* OR limitation* OR access OR guideline*)) OR ("psychoacoustic" AND "spectrotemporal" AND evaluation)* – focused on cochlear implant candidacy, referral processes, associated barriers and psychoacoustic evaluation methods, combining clinical and psychoacoustic perspectives. Despite these advances, underutilization of CIs has persisted, highlighting an ongoing gap between eligibility and referral that necessitates improved decision-support approaches. The second search string – *("clinical decision support" OR CDS OR "decision aid") AND ("design guidelines" OR "design criteria" OR "design characteristics" OR "interface design")* – therefore addressed CDS design, focusing on design guidelines and interface considerations. Results were limited to peer-reviewed, English-language publications focusing on adult populations from 2010 onwards, reflecting contemporary clinical practices and expanded CI candidacy criteria (Firszt et al., 2012; Távora-Vieira et al., 2013).

Screening was conducted in two stages: First, titles and abstracts were reviewed; second, full texts were assessed for relevance, resulting in 47 publications. Studies were excluded if they focused on highly specific conditions, device-level technical aspects, pediatric populations, post-implant care or domains unrelated to candidate identification or CDS design. Following the recommendations of Webster and Watson (2002), the search process was iterative, incorporating backward and forward searching, and continued until theoretical saturation was reached. In total, a final set of 100 publications was included and organized into thematic sections forming the foundation of this research.

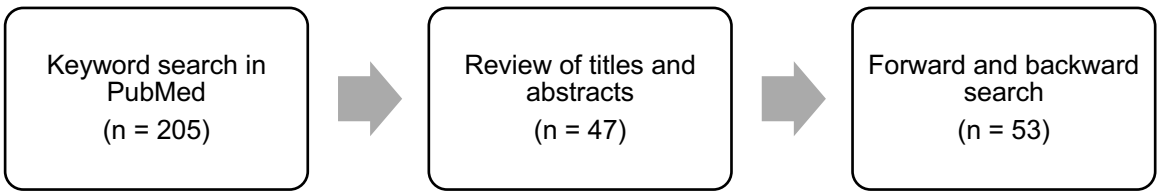


Figure 1. The structured literature review process

3.2 Primary Data Collection

The empirical data for this study were collected through a self-administered online questionnaire. It was informed by the preceding literature review, which established a conceptual framework and identified key themes, and guided by established principles outlined by Clark et al. (2021) and Ranganathan and Caduff (2023). This method was chosen due to its practical advantages, particularly for geographically dispersed participants. As the target group consisted of clinicians across Sweden, the online format enabled efficient data collection with minimal time and cost, while allowing respondents to complete the questionnaire flexibly alongside their professional

responsibilities. Compared to interviews, this reduced the burden on participants and lowered the barrier to participation.

Given the absence of an interviewer, particular attention was paid to the design of clear and unambiguous questions. Questions were carefully formulated to avoid bias and misinterpretation and were reviewed internally with a MED-EL colleague. A pilot study was not conducted due to time constraints. The questionnaire was kept concise to reduce respondent burden and combined close-ended, hybrid and open-ended formats. Closed-ended questions enabled efficient analysis and facilitated comparison, while hybrid and open-ended questions allowed for flexibility and the capture of unanticipated insights. All questions were mandatory to minimize incomplete responses.

The survey was implemented using Microsoft Forms via a university account, which complies with GDPR requirements and provides a user-friendly interface that does not require participant registration. An introductory section outlined the study purpose, data privacy information and contact details, followed by the survey questions and a consent confirmation.

3.3 Participants

Participant recruitment was closely linked to the empirical data collection strategy described above. As self-administered questionnaires are commonly associated with lower response rates and limited control over sampling, recruitment was conducted through MED-EL to mitigate these challenges. This approach enabled access to a highly relevant target group and enhanced the study's credibility. In addition, distribution through MED-EL's established professional networks increased the likelihood that the questionnaire reached the intended participants and reduced the risk of responses from individuals outside the target group. Through this collaboration, MED-EL facilitated contact with a clinician involved in cochlear implantation at Karolinska Institutet in Stockholm. The questionnaire was initially distributed to audiologists during an internal informational meeting and subsequently shared via email as a follow-up within this professional network. To reduce the risk of undue influence and potential bias related to the industry collaboration, no direct contact was established between the researcher and the participants. The inclusion criteria specified clinicians, medical students and individuals with relevant experience related to cochlear implantation or hearing care in Sweden, as the study aimed to capture perspectives on the early stages of the CI referral pathway. Participation was voluntary, and no remuneration was provided.

In total, four participants completed the questionnaire. All respondents indicated that they were based in Sweden. The sample consisted of three audiologists working within primary care settings and one bimodal CI user who had personally undergone the referral process. The small, specific sample, along with reliance on voluntary participation, introduces potential volunteer bias and limits the representativeness of the findings. These limitations are acknowledged, and the results are therefore interpreted as exploratory insights intended to inform the design and development of the proposed CDS concept rather than generalizable conclusions.

3.4 Data Analysis

The analysis followed a qualitative approach, as the study primarily relied on textual data from open-ended questionnaire responses. Close-ended questions were used to establish contextual background which complemented and informed the interpretation of the qualitative data.

A thematic analysis was conducted, providing a structured yet flexible approach to identifying, analyzing and interpreting patterns of meaning across the dataset. The analysis adopts a contextualist stance, in which themes are grounded in participants' accounts but understood as context-dependent and co-constructed through the interaction between participants' responses and the researcher's interpretation. Accordingly, the analysis not only reflects clinicians' experiences but also interprets them in relation to the broader clinical context and translates them into design implications. The analysis was guided by the six-phase framework proposed by Braun and Clarke (2006).

- In the first phase, familiarization with the data, all responses were read repeatedly to gain an overall understanding of the content and to identify initial patterns and impressions. This step enabled immersion in the data and formed the basis for subsequent analysis.
- In the second phase, during the generation of initial codes, relevant semantic segments of the data were systematically coded.
- The third phase involved searching for themes, in which related codes were grouped together to form broader patterns of meaning.
- In the fourth phase, reviewing themes, the identified themes were refined by examining their coherence and relevance to both the coded data and the overall dataset, ensuring they accurately represented the underlying data.
- In the fifth phase, defining and naming themes, each theme was clearly articulated in terms of its core meaning and contribution to the research question, ensuring clarity and distinction between themes.
- Finally, in the sixth phase, writing the report, the themes were synthesized into a coherent narrative and interpreted within the context of the case study. These interpretations were then used to inform and iteratively translate the findings into the design of a CDS prototype.

3.5 Ethical Considerations

This study was conducted in accordance with established principles of good research practice, drawing on the guidelines outlined by the Swedish Research Council (Vetenskapsrådet, 2024) based on the European Code of Conduct for Research Integrity (ALLEA - All European Academies, 2023). Particular attention was given to ensuring that participation was informed, voluntary, specific, explicit and unambiguous, possible to withdraw and appropriately documented (Vetenskapsrådet, 2024).

- Participants received clear information about the study's purpose, its context within a master's thesis in collaboration with MED-EL, and the handling of data. The questionnaire was fully anonymous, with no personal data collected, and results were reported only in aggregated form. The researcher's contact details were provided for further inquiries.
- Participation was entirely voluntary, with no incentives or pressure to participate. Indirect recruitment through a clinician who was not affiliated with MED-EL further reduced the risk of undue influence, as participation was entirely at the discretion of the respondents.
- Consent was specific, as participants were clearly informed about the study's purpose, their role, and how their responses would be used within this defined research context.
- Consent was explicitly obtained via a required checkbox prior to submission, confirming understanding and agreement to the use of responses for research purposes.

- Participants could withdraw at any time before submission by exiting the questionnaire. However, withdrawal after submission was not possible due to the data's anonymity.
- Consent was documented through the digital confirmation within the questionnaire, ensuring that participants made an informed and voluntary decision to take part.

4 Findings

Following the coding process, the findings chapter is structured around two main themes identified through the thematic analysis: Current Assessment Practices and Multifactorial Influences on Referral Decision-Making and Conditional Acceptance of Clinical Decision Support. The coded responses were interpreted and synthesized into a coherent narrative within the context of the case study, forming the foundation for the interpretative translation of the findings into a clinician-facing CDS interface design presented in Section 4.3.

4.1 Current Assessment Practices and Multifactorial Influences on Referral Decision-Making

Across the audiologists, referral decisions were primarily based on pure-tone audiometry, word recognition with earphones and patient self-report, indicating a strong reliance on standard and readily available assessment methods that are not specifically designed for CI candidacy assessment. Patient-reported difficulties in everyday listening situations, particularly despite HA use, appeared to play an important complementary role alongside audiological measures. One participant also reported using aided word recognition testing, suggesting that more specialized assessments associated with CI candidacy evaluation may not be routinely incorporated into all referral-related assessments. The bimodal CI user similarly reported that both pure-tone audiometry and self-reported hearing difficulties played important roles in the referral process. In addition, participants indicated that factors such as age, general health status and expectations regarding rehabilitation may also influence whether patients are referred for specialized CI evaluation after initial assessment.

Among the three audiologists, one participant reported rarely referring patients with hearing loss who may be suitable for cochlear implantation for further evaluation, while two stated that they always do. Similarly, responses regarding the perceived difficulty of referral decision-making ranged from very difficult to neutral and very easy, while self-reported confidence ranged from not so confident to quite confident and very confident. Although no clear pattern can be identified within this small sample, the variation itself reflects differences in experience, confidence and approaches to referral decision-making. Importantly, all participants indicated that they had encountered patients they believed could have benefited from cochlear implantation, but who were ultimately not referred for further evaluation, highlighting gaps and missed opportunities in current referral pathways. While some participants associated referral uncertainty with factors identified in the literature review – including unclear candidacy criteria, uncertainty regarding CI benefit and a lack of clear referral guidelines – two of the three audiologists additionally emphasized patient reluctance as a major reason why referral may not occur, suggesting that barriers to referral are not solely clinician- or system-related, but may also involve patient perceptions, concerns and expectations regarding cochlear implantation and rehabilitation. In addition, the limited scope of

reported assessment methods may point toward restricted access to more comprehensive testing procedures in primary care.

Overall, the findings indicate that referral decision-making occurs within complex, variably constrained clinical environments, where structural limitations, subjective interpretation and patient-related factors may influence whether potential CI candidates proceed to specialized evaluation.

4.2 Conditional Acceptance of Clinical Decision Support

While all participants expressed overall openness toward the use of a self-administered hearing screening tool, the findings indicate that acceptance depends on specific conditions related to usability, workflow compatibility, interpretability and flexibility. All participants stated that they would use such a tool if it supported the identification of potential CI candidates without increasing workload, highlighting efficiency and time requirements as central prerequisites for adoption. Participants' views regarding where such a tool would be most useful within the patient journey varied considerably. One audiologist indicated that it would be most useful before the consultation as a first indication, one preferred its use during the consultation as part of the assessment process, and one saw the greatest value after the consultation to support referral decisions. This variation suggests that clinicians may require different forms of integration depending on their workflows and clinical contexts, underscoring the need for adaptability and flexibility in the system design.

The CI user additionally emphasized the value of a language-agnostic approach, stating that such a design removes a major barrier for patients with limited language proficiency or cognitive impairment. The participant further highlighted that the tool should ideally be usable independently by patients before appointments and that results should support meaningful conversations between clinicians and patients rather than replace them. Reflecting on personal experiences with delayed referral, the participant stressed that early identification is critical, as prolonged under-referral can have substantial consequences for quality of life, work capacity and mental health.

Preferences for result presentation also varied considerably, further supporting the need for a flexible, customizable tool. Among the clinicians, participants expressed different preferences for a traffic-light indicator of the likelihood of suitability for CI referral, a written summary of results, a visualized hearing profile and comparison with normal hearing ranges. Each clinician showed a relatively clear preference toward one of the presentation formats. The bimodal CI user, in contrast, preferred a combination of a visualized hearing profile, a traffic-light-style suitability indicator and a written explanation of the results, suggesting a greater need for contextualization and layered information. Similarly, preferences regarding information depth varied. While most participants expressed a preference for brief, simplified results focusing only on key factors, one audiologist preferred more detailed explanations including underlying data and reasoning. At the same time, all participants emphasized the importance of clear explanations of how results are generated and the ability to review underlying reasoning, highlighting transparency as a central prerequisite for trust in such systems. Ease of use, rapid interpretation, integration with existing systems and minimal time requirements were also repeatedly identified as important conditions for adoption. Participants additionally reported using a wide variety of digital systems for patient data management, including Webdoc, Otosuite and TakeCare, reflecting the fragmented nature of existing clinical infrastructures. While only one clinician explicitly stated that automatic integration of screening results into EHR systems was essential, two clinicians and the CI user identified EHR-based

documentation as the most convenient option for handling results, whereas one participant preferred printable reports.

Together, these findings suggest that future implementations should remain adaptable across different documentation practices and technical environments rather than assuming a single standardized workflow. Taken together, this theme indicates that acceptance of decision-support tools depends not only on technological availability but also on whether such systems can integrate flexibly into diverse clinical workflows while remaining efficient and adaptable to varying user preferences and patient needs.

4.3 Interpretative Interface Design for CDS

The findings highlighted that clinicians' willingness to use such systems depends strongly on workflow compatibility, rapid interpretation, ease of use and transparency and adaptability to different clinical contexts and user preferences. At the same time, participants expressed highly varied preferences regarding when such a tool should be used within the patient journey, how results should be presented and how much detail should be displayed. These findings reinforced that the prototype could not assume a single standardized workflow, interaction style or information preference. Instead, the system needed to remain flexible enough to support different uses while minimizing interpretation effort and cognitive load. These findings directly informed the interface, visualization and interaction principles implemented throughout the prototype. To address these requirements, the first results screen was intentionally designed as a simplified and low-threshold overview requiring minimal interpretation effort (see Figure 2).

A visually simplified concentric infographic was introduced to provide rapid initial orientation before users access more detailed information. The visualization is based on eight difficulty levels representing increasingly complex listening conditions, with performance normalized against reference groups to position results along a continuum ranging from listening difficulty (1) to listening ease (8), thereby facilitating comparison across different hearing levels. This design decision was directly informed by participants who preferred brief and simplified visual outputs focused primarily on key factors.

At the same time, the concentric visualization mirrors the visual logic of the more detailed radar chart presented in the second interface layer, where larger outer areas indicate better performance in both representations (see Figure 4). This consistency was intended to reduce cognitive switching between overview and detailed analysis while supporting quick visual interpretation. The design was partly inspired by the traffic-light style suitability indicators preferred by some participants and previously explored in the earlier prototype iteration. However, because participants also emphasized the importance of transparency and nuanced interpretation, the visualization was expanded beyond a simple three-level categorization into a graded continuum representing several levels of listening difficulty. In addition to a concentric visualization that represents the combined performance of both ears as a single overall difficulty level, a second visualization separates results for the left and right ears to provide clinicians with greater interpretative flexibility during assessment and diagnosis (see Figure 3).

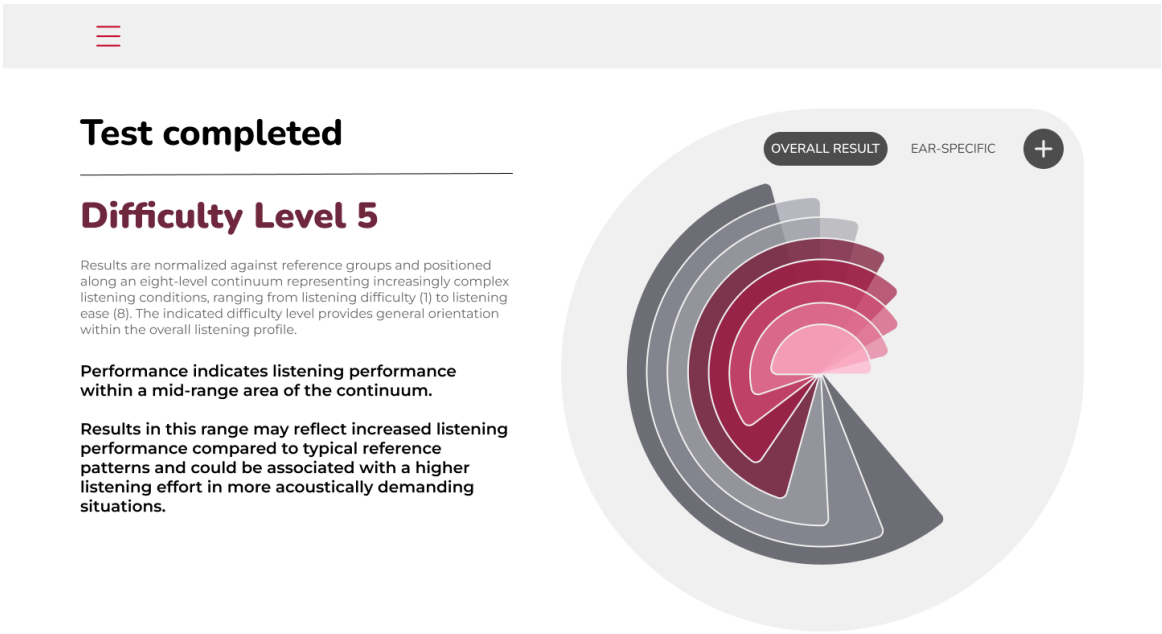


Figure 2. General overview visualization showing the combined overall listening difficulty level across an eight-level continuum, ranging from listening difficulty to listening ease

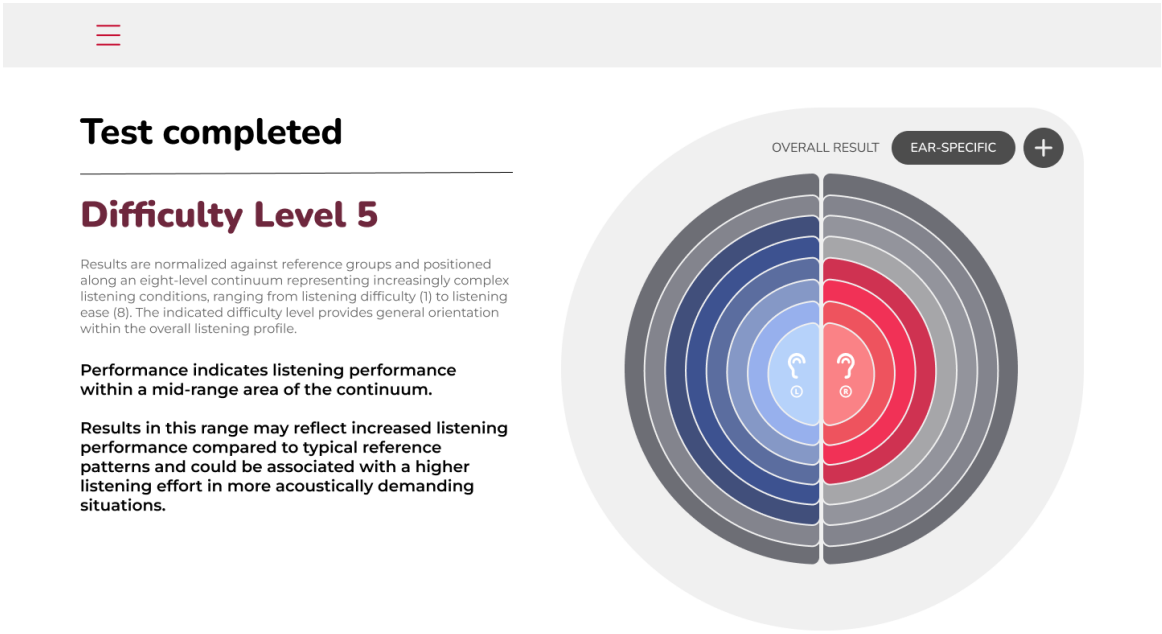


Figure 3. An ear-specific overview visualization displaying separate listening difficulty levels for the left and right ear using established audiological color conventions

At the same time, the visual design needed to align with established conventions within audiological practice. In audiological settings, red and blue are commonly used to represent the right and left ears respectively. Maintaining this color consistency throughout the interface was therefore considered important in order to align the prototype with clinicians' existing perceptual habits and expectations. While the initial visualization concept relied more strongly on traditional traffic-light colors such as red, yellow and green to indicate different levels of difficulty, this approach created potential ambiguity within the audiological context, where red already carries a specific meaning. Accordingly, red and blue were used consistently for ear-specific results, while purple was introduced to represent combined performance. Softer gradients and more nuanced color transitions were introduced as alternatives to classic traffic-light indicators to preserve rapid visual orientation while avoiding overly deterministic interpretations, which could conflict with the intention to position the prototype as a supportive and non-directive CDS concept.

The findings also showed that users differ considerably in the amount of contextual information they prefer. While most participants favored concise summaries, others requested access to more detailed explanations, underlying reasoning and contextually relevant result interpretation. The second visualization layer therefore provides a more detailed radar chart displaying psychoacoustic dimensions, allowing users to explore how individual dimensions contribute to the overall listening profile (see Figure 4). The expanded view is accessed by clicking the plus icon, enabling users to progressively move from a simplified overview to a more detailed interpretation. Rather than functioning as a black-box recommendation system, the prototype supports clinician autonomy by making the basis of the result interpretable and explorable, while also extending beyond the presentation of raw data by providing contextually relevant and evidence-informed orientation. In this sense, the prototype also serves as micro-learning, exposing clinicians to psychoacoustic dimensions that are not routinely incorporated into conventional referral-related assessments.

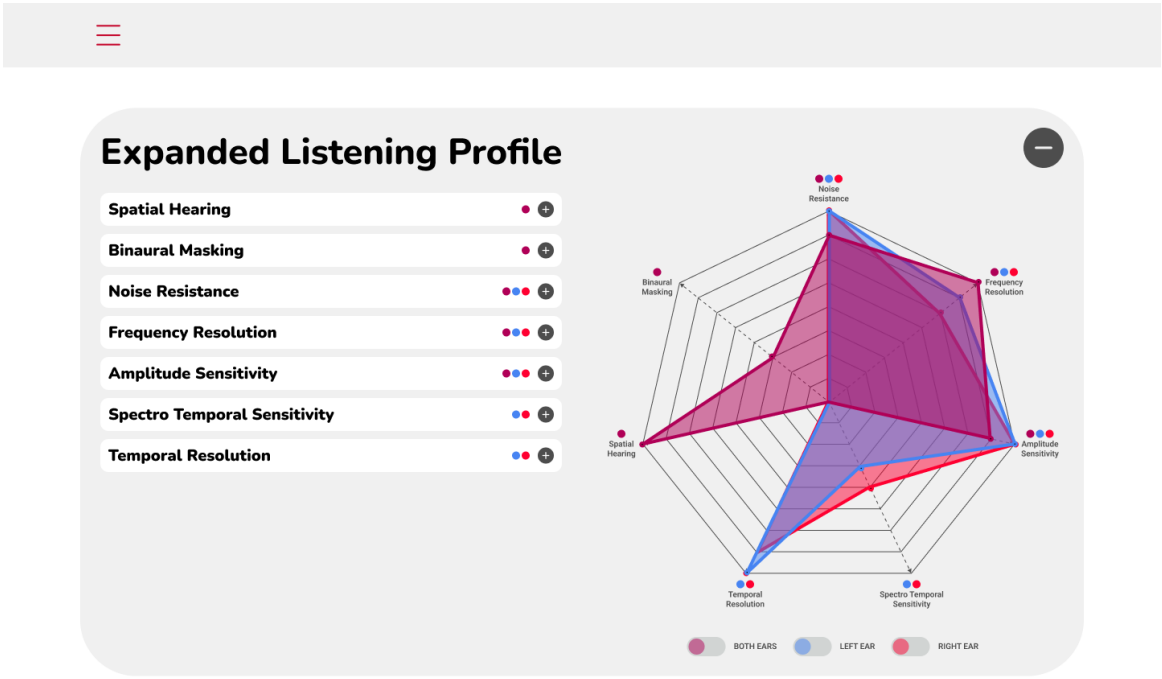


Figure 4. Expanded listening profile displaying combined screening results within a radar-chart visualization

To support interpretability without overwhelming users, expandable interface components were integrated throughout the design. Expandable menus allow clinicians to initially focus on core information while optionally accessing more detailed explanations or references to clinical evidence when needed, reflecting both participant preferences for layered information and CDS literature recommending progressive disclosure mechanisms to reduce cognitive load.

Because some psychoacoustic tests are conducted bilaterally while others are performed separately for each ear, the radar chart is not always fully populated. To improve readability and preserve flexibility, hover interactions were designed to visually bring the selected result dimension to the foreground while greying out less relevant information. This allows clinicians to focus more closely on unilateral versus bilateral results by reducing visual complexity. Clinicians can additionally toggle individual results on and off depending on the interpretation context (see Figure 5). In contrast to the hover interaction, toggling actively restructures the radar chart itself by removing irrelevant dimensions rather than merely greying them out, allowing the remaining visualization to appear as a complete shape. This was intended to support more natural and intuitive interpretation of individual result profiles while simultaneously reducing visual clutter and accommodating different user preferences and clinical use cases.

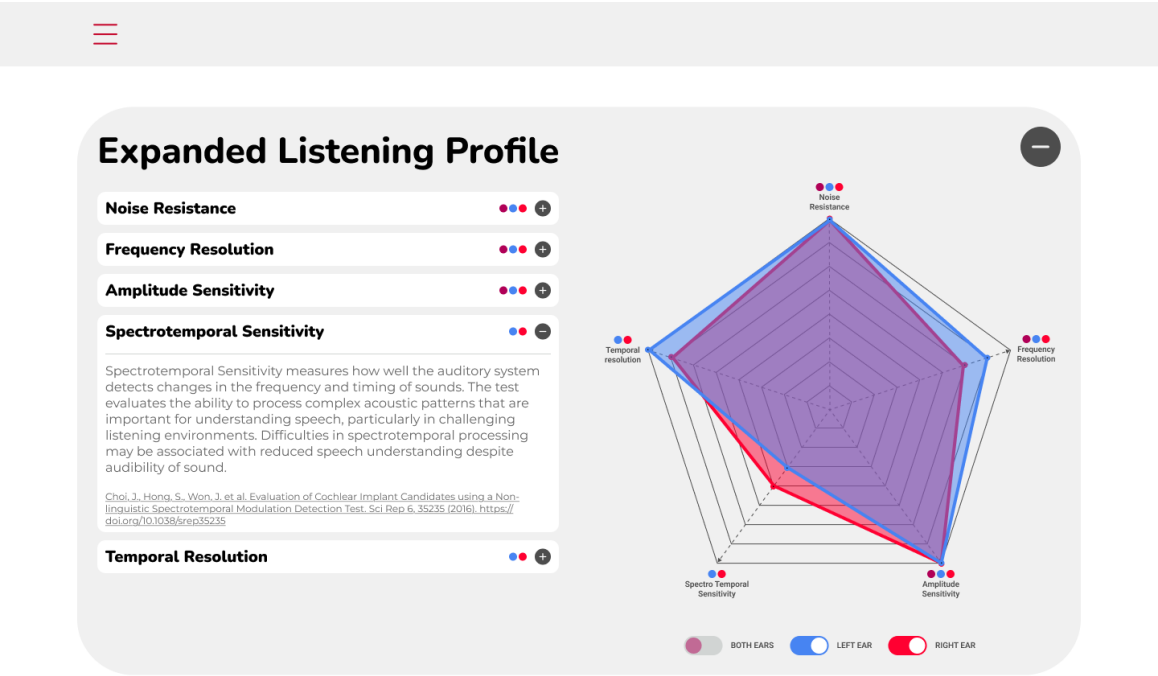


Figure 5. Customized radar-chart view with selectable result views, accompanied by brief test descriptions and additional explanatory resources

5 Discussion

Building on an earlier internship project conducted in collaboration with MED-EL, this thesis extends a previously developed self-screening concept into a clinician-facing CDS context. The original project focused on designing an engaging, research-based and user-centered hearing

screening experience intended to raise awareness of hearing loss and CIs among lay users while supporting early orientation regarding potential hearing difficulties. Developed through an iterative two-month design process involving research, prototyping and interdisciplinary collaboration, the concept emphasized storytelling and approachable communication rather than clinical assessment, resulting in an interactive, gamified prototype that framed hearing screening as a positive and empowering step toward improved hearing well-being with the potential for cochlear implantation. While the test flow itself remained unchanged and is therefore not the primary focus of this thesis, the project established the conceptual and design foundation upon which the present research builds by exploring how psychoacoustic screening results could be further contextualized and visualized to additionally support clinicians during early CI referral decision-making. In doing so, the study positions itself at the intersection of hearing healthcare, psychoacoustic research, CDS design and regulatory considerations surrounding digital health technologies.

The literature review and empirical findings jointly reinforced many of the structural and diagnostic challenges associated with current CI referral pathways and further strengthened the validity and relevance of exploring psychoacoustic approaches as a potential support mechanism for early CI referral decision-making. Together, the findings reinforce the growing body of literature suggesting that under-referral for cochlear implantation is not the result of insufficient clinical effectiveness or narrow candidacy criteria, but is embedded within structural, organizational and perceptual barriers throughout the hearing-care pathway, making them difficult to address through short-term organizational changes alone. While candidacy criteria and evidence of benefit continue to expand, referral practices have not evolved at the same pace. Consistent with previous research, participants relied primarily on conventional, readily available assessment methods such as pure-tone audiometry, word recognition under earphones and patient self-report when making referral-related decisions.

At the same time, participants acknowledged that referral decisions may additionally be shaped by subjective assumptions regarding rehabilitation expectations, patient reluctance, assumptions regarding expected benefit and contextual factors extending beyond purely audiological indicators, although they may be more appropriately addressed at later stages of the evaluation process rather than during the initial identification and referral stage. Importantly, all participants reported encountering patients they believed could have benefited from cochlear implantation but who were ultimately not referred for specialized evaluation. Although the sample size was small and no generalized conclusions can be drawn, this finding aligns closely with the broader literature describing persistent under-identification and under-referral of potential CI candidates. The findings therefore support the argument that current referral pathways may not consistently facilitate timely identification of suitable candidates, particularly in contexts where clinicians lack access to specialized testing procedures and where uncertainty regarding CI candidacy criteria or patient reluctance toward cochlear implantation and rehabilitation may further complicate referral decisions.

In this context, the study further strengthens the relevance of psychoacoustic approaches as a potentially valuable support mechanism during earlier stages of the referral pathway. While best-aided speech recognition testing remains the clinical gold standard for candidacy evaluation, the literature review highlighted several practical limitations associated with its implementation, including extensive fitting requirements, language dependency, specialized equipment and controlled testing environments. Rather than attempting to redesign referral infrastructures directly, the prototype therefore focuses on exploring a more immediately implementable and accessible support mechanism that can operate within existing clinical constraints. Psychoacoustic measures

therefore appear particularly promising because they are speech-agnostic, comparatively efficient and less dependent on highly specialized clinical infrastructures. The findings of this study support the relevance of these characteristics, as clinicians emphasized the importance of workflow compatibility, rapid interpretation and low additional workload when considering whether to adopt a CDS tool in practice.

It should be noted, however, that the development and validation of the test procedures are still ongoing. Consequently, both the specific test procedures presented within the prototype and the number of included tests should be understood as provisional and primarily intended to visualize the overall CDS concept rather than represent a finalized assessment battery. Similarly, much of the textual content, labels and explanatory descriptions included in the interface currently serve illustrative and conceptual purposes within the prototype and require further refinement and validation in future development stages.

5.1 Translating Clinical Needs into a CDS Concept

At the same time, the study highlights that successful CDS adoption is not merely a technological question, but strongly dependent on how such systems align with the realities of clinical practice. Participants expressed highly varied preferences regarding when a CDS tool would be most useful within the patient journey, ranging from pre-consultation screening to consultation support and post-consultation referral orientation. Similarly, preferences regarding information presentation differed considerably, with some participants preferring minimal, rapidly interpretable outputs, while others emphasized the need for contextual explanations, visualized hearing profiles and access to the underlying reasoning. These findings reflect the broader CDS literature which emphasizes that usability, interpretability, workflow integration and clinician autonomy are critical determinants of adoption. Rather than supporting the idea of a single standardized solution, the findings suggest that future CDS systems within hearing healthcare may need to remain adaptable across different workflows, user preferences and clinical environments.

These findings directly informed the prototype's design rationale. From a design perspective, the study contributes by demonstrating how empirical findings can be translated into concrete interface and interaction principles within a CDS context. The prototype intentionally combines low-threshold orientation with layered interpretability. Simplified concentric visualizations, graded listening difficulty indicators and radar-chart representations were designed to support rapid sense-making while still enabling deeper exploration of psychoacoustic dimensions and underlying reasoning when needed. Progressive disclosure mechanisms, hover interactions and customizable radar-chart filtering were implemented specifically in response to participants' preferences for flexibility, transparency and variable levels of detail. In this sense, the prototype represents an interpretative translation of psychoacoustic research and CDS design principles into a clinician-facing concept intended to support orientation during CI referral decision-making.

The inclusion of a bimodal CI user additionally provided an important complementary perspective that extended beyond purely clinical considerations. While this study focused primarily on clinicians, participants' reflections highlighted that CDS systems operating within referral pathways also ultimately shape patient experiences. The participant emphasized the importance of language-agnostic design, independent usability and meaningful communication between clinician and patient. This perspective is particularly relevant given that patient reluctance itself emerged as a potential barrier to referral within the findings. The prototype therefore intentionally avoids

functioning as an opaque recommendation system and instead emphasizes interpretability to support shared orientation and discussion rather than automated decision-making. Nevertheless, because the prototype requires clinical interpretation and presents information in ways that could potentially be misinterpreted by non-professional users, the current concept remains primarily intended for supervised clinical use. Future iterations would likely require adaptation before independent patient-facing deployment could be considered appropriate.

5.2 Regulatory Tensions in CDS Design

The findings also highlight a broader conceptual tension increasingly characterizing digital health technologies: the balance between providing clinically meaningful support, avoiding over-automation and preserving clinician autonomy. This tension became particularly relevant in light of regulatory considerations surrounding the European Medical Device Regulation (MDR). Because software intended to diagnose or directly recommend treatment may qualify as a medical device, it can require extensive clinical validation, conformity assessment and regulatory approval processes, significantly increasing development complexity and time to deployment (Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on Medical Devices, 2017). Consequently, the design process was strongly shaped by the intention to position the prototype within a non-regulated CDS space that emphasizes transparency, interpretability and clinician autonomy rather than automated decision-making. Rather than generating referral recommendations, the prototype was therefore intentionally designed as a supportive, non-directive orientation tool. This approach aligns not only with CDS literature emphasizing the importance of explainability and clinician autonomy, but also with concerns regarding trust, accountability and overreliance on opaque algorithmic systems in healthcare environments (Fenech et al., 2018).

However, the boundary between regulated and non-regulated software remains difficult to define and is not always clear, particularly for digital tools such as decision support systems, self-monitoring applications and EHRs (Läkemedelsverket, 2025). Although the MDR was introduced in response to the rapidly evolving digital health landscape to support safe innovation, classification depends heavily on the defined intended use and its interpretation by regulatory bodies, making the process to some extent subjective (Van Laere et al., 2021). Consequently, the prototype remains interpretative not only in its translation of literature and empirical findings into interface and interaction design decisions, but also in its assumed positioning within a non-regulated CDS context. This ambiguity reflects broader challenges in the rapidly evolving digital health landscape, where technological innovation often advances faster than regulatory clarification (*Ethics and Governance of Artificial Intelligence for Health*, 2021).

5.3 Limitations and Future Research

Nevertheless, several limitations must be acknowledged. Most importantly, the study was based on a very small and highly specific sample. While this limitation was not considered critical given the research's exploratory nature, it substantially limits the generalizability of the findings. At the same time, the difficulty of recruiting appropriate participants itself reflects the broader challenge of accessing specialized clinical populations within constrained research timeframes. Furthermore, because the study relied on qualitative interpretation and exploratory prototyping rather than clinical validation, the proposed concept cannot be understood as evidence of diagnostic effectiveness or clinical utility. Instead, the findings should be understood as an initial exploratory foundation informing future iterative development.

Another limitation concerns the limited availability of research specifically addressing CDS interface design within CI referral contexts. As a result, the prototype design relied heavily on broader CDS literature and interpretative synthesis rather than domain-specific interface guidelines. In addition, the narrative nature of the literature review introduces a degree of subjectivity regarding study selection, synthesis and design translation, reflecting broader limitations frequently acknowledged within qualitative and design-oriented research.

Against this backdrop, the next step should focus on iterative, practice-based evaluation of the prototype in real clinical environments. In particular, UCD approaches such as Action Design Research (ADR) appear highly relevant, as they would allow the prototype to be continuously refined through active implementation, evaluation and collaboration with clinicians and patients in practice. Such approaches would additionally provide opportunities to evaluate usability, workflow integration, interpretability and real-world clinical impact over time. More extensive qualitative approaches, particularly interviews and longitudinal evaluations, could further provide richer insight into how clinicians and patients experience such systems in practice, how trust is established and how workflow integration influences long-term adoption. At the same time, future studies should investigate how such systems can be better adapted for independent patient-facing contexts without increasing the risk of misinterpretation or anxiety. While the current prototype intentionally prioritizes concise and rapidly interpretable information presentation, particularly through brief explanations and expandable dropdown menus within the detailed results view, future iterations could further expand these informational layers. More extensive explanatory views or dedicated educational screens could provide clinicians and patients with deeper insight into the psychoacoustic test methods, interpretation of results and underlying reasoning when desired. In this sense, the prototype also presents opportunities for integrated learning during use.

Future development should additionally explore deeper integration with EHR systems and adaptive visualization approaches that support different forms of clinical interpretation and workflow integration. Although integration into EHRs was identified as important for long-term clinical usability, this aspect was beyond the scope of the current prototype. However, this appears particularly relevant given the current changes to Sweden's healthcare infrastructure. Several Swedish regions are in the process of replacing their journal systems. Within the coming years, multiple regions are expected to adopt Cambio Cosmic, while major regions, Skåne and Västra Götaland, are instead investigating alternatives (Angvarson Valberg, 2025; Beslut i regionstyrelsen, 2026; Cederberg, 2023). Since responses regarding currently used EHR systems were already highly scattered within this small sample, future research should investigate which systems should be prioritized for interoperability and implementation in order to support broader clinical applicability.

6 Conclusion

This thesis set out to answer the question: *How can a self-administered hearing screening tool be designed to support clinicians in the early identification of potential cochlear implant candidates?*

By combining insights from psychoacoustic research, CDS literature, empirical findings and regulatory considerations, the study demonstrated that such a system may hold value not as a diagnostic replacement but as a supportive and interpretable orientation tool within existing referral

pathways, adding to the field of Informatics and Digital Service Innovation. The findings indicate that current CI referral practices remain shaped by structural limitations, fragmented workflows, uncertainty regarding candidacy and reliance on conventional assessment methods that are not specifically designed for CI referral decision-making. At the same time, clinicians emphasized the importance of rapid interpretation, workflow compatibility, transparency and flexibility when considering the adoption of digital support tools. These findings directly informed the development of a clinician-facing CDS concept designed around low-threshold orientation, layered interpretability and clinician autonomy. Rather than generating automated referral recommendations, the prototype explored how psychoacoustic screening results can be contextualized and visualized in ways that support clinical judgment while remaining understandable, accessible and adaptable to different clinical contexts.

The study therefore suggests that the potential value of psychoacoustic screening within CI referral pathways may lie less in replacing existing clinical assessment practices than in complementing them through accessible, workflow-sensitive forms of support. In this sense, the proposed concept represents a complementary pathway for supporting earlier CI candidate identification within existing clinical constraints. Rather than attempting to redesign referral infrastructures entirely, the study demonstrates how transparent, non-directive and human-centered CDS concepts may help bridge the gap between emerging candidacy research and routine clinical practice.

Ultimately, the thesis highlights that improving CI referral is not solely a technological challenge, but also a design challenge. In doing so, the study contributes not only to discussions surrounding CI referral and psychoacoustic screening, but also to broader conversations regarding the role of transparent, non-directive and human-centered CDS systems within digital healthcare. Supporting earlier identification depends not only on what information is available, but on how that information is communicated, interpreted and integrated into the realities of clinical decision-making.

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